

Deterministic Optimization

Discrete Optimization:
Modeling w/ binary
variables

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Modeling Exercises 3

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Learning Objectives

- Observe modeling a mixed integer **nonlinear** programming problems

Portfolio Optimization

- We are given a set of n stocks
- The expected return (in \$/share) of stock j is estimated from historical data and is given by r_j
- The covariance between the return of asset i and j is σ_{ij}
- We want to determine how many shares of each stock to buy so as to minimize risk (variance) subject to some investment constraints

Portfolio Optimization: Constraints

- We can invest in at most k stocks (here $k < n$)
- The expected return of the portfolio should be at least R
- If we invest in stock j we must buy at least l_j shares
- We cannot buy more than u_j shares of stock j

Formulation

Variables:

x_j : Number of shares of stock j to buy (nonnegative integer)

y_j : Whether to invest in stock j (binary)

$$\begin{aligned} \min \quad & \sum_{i=1}^n \sum_{j=1}^n \sigma_{ij} x_i x_j \\ \text{s.t.} \quad & \sum_{j=1}^n r_j x_j \geq R \\ & l_j y_j \leq x_j \leq u_j y_j \quad \forall j = 1, \dots, n \\ & \sum_{j=1}^n y_j \leq k \\ & x_j \in \mathbb{Z}_+, \ y_j \in \{0, 1\} \quad \forall j = 1, \dots, n \end{aligned}$$

Remarks

- The formulation is an example of a mixed integer (convex) quadratic program (MIQP)
- Dropping integrality we get a relaxation that is a (convex) quadratic program
- Similar formulation arises in sparse regression applications
- Such formulation can be solved by solvers such as `cplex` or `gurobi`

Summary

- We observe an example of a mixed integer nonlinear program